

CS 112

Introduction to Data Structures

WEEK 04

PyList: Dynamic Arrays

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This Week

- ➊ Why a plain array isn't enough
- ➋ `new` and `delete`
- ➌ The stack and the heap
- ➍ Building a dynamic array
- ➎ Growing when full
- ➏ Destructors: now you need one
- ➐ Copy constructors: shallow vs deep **KEY**

Arrays: The Good and the Bad

Good



- Simple to declare
- Indexing is fast: one multiply and add
- Contiguous, so the cache loves them

Bad



- Size fixed at **compile time**
- Can't grow when you need one more
- No idea how many slots are actually used

Python's `list` grows on demand. This week we build that, and find out what it costs.

Memory You Ask For

```
int *arr = new int[capacity]; // ask the OS for space

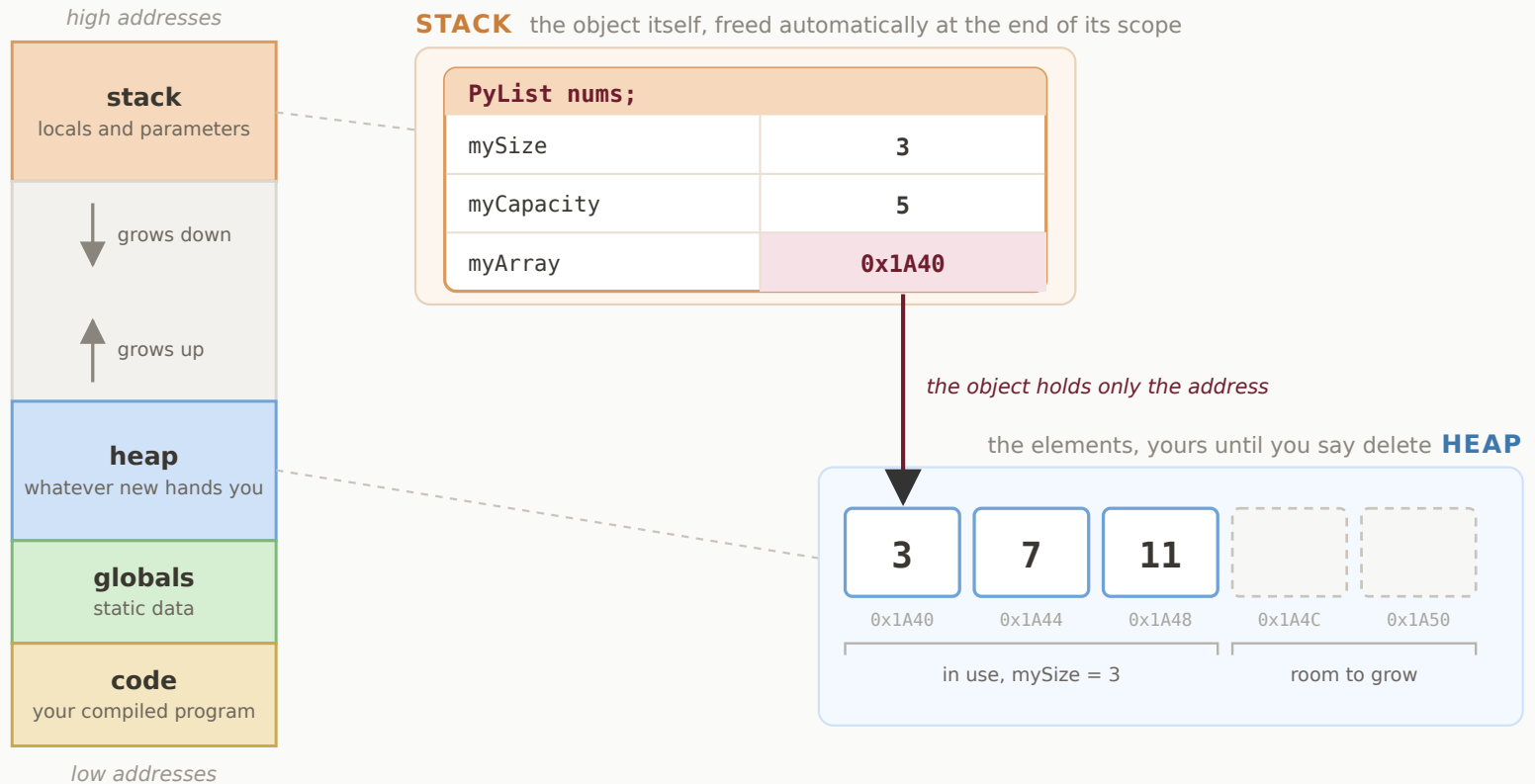
arr[0] = 42;                  // use it like an array

delete [] arr;                // give it back
arr = nullptr;                // don't keep the address
```

- `new` returns the **address** of the block
- The size can be a variable: decided while the program runs
- `delete []` for arrays, plain `delete` for single objects

The `new` command needs a matching `delete`. So if you create new memory, you have to remember to free the space afterwards.

Stack and Heap



The object goes away by itself at the end of its scope.
The block it points at does not: that one is yours to free.

A Dynamic Array

```
class PyList {
public:
    PyList();
    PyList(const PyList &original);           // copy constructor
    ~PyList();                               // destructor
    unsigned getSize() const;
    void      append(const Item &it);
    Item&      operator[](unsigned i);
private:
    Item      *myArray;                      // heap block
    unsigned  mySize;                        // slots in use
    unsigned  myCapacity;                    // slots available
};
```

size is how many items you have. **capacity** is how many fit before it must grow. Keeping them apart is what makes appending cheap most of the time.

Appending When There's Room

```
void PyList::append(const Item &it) {  
    myArray[mySize] = it;  
    ++mySize;  
}
```

before [3 | 7 | _ | _] size 2, cap 4

after [3 | 7 | 11 | _] size 3, cap 4

Two operations, no matter how big the array is. **Constant time.**

Appending When It's Full

```
void PyList::append(const Item &it) {
    if (mySize == myCapacity) {
        unsigned cap = (myCapacity == 0) ? 1 : myCapacity*2;
        Item *bigger = new Item[cap];
        for (unsigned i = 0; i < mySize; ++i) {
            bigger[i] = myArray[i];           // copy everything
        }
        delete [] myArray;                   // release the old
        myArray = bigger;
        myCapacity = cap;
    }
    myArray[mySize] = it;
    ++mySize;
}
```

Now it copies all n items: **linear time**. Doubling means this happens rarely, more on that next week.

TALK TO YOUR NEIGHBOR · TTYN

Why double the capacity instead of adding one slot?

- A.** Doubling uses less memory
- B.** Adding one would make every append copy the whole array
- C.** The heap only hands out sizes that are powers of two
- D.** It doesn't matter, both are equally fast

Now You Need a Destructor

```
PyList::~~PyList() {  
    delete [] myArray;  
    myArray = nullptr;  
}
```

C++ writes a destructor for you, but it only destroys the *members*. It destroys the pointer, not what the pointer points at.

The rule: if your class calls `new`, your class must define a destructor. Otherwise every `PyList` that dies leaves its heap block behind.

TALK TO YOUR NEIGHBOR · TTYN

When does a destructor run?

- A.** Only when you write `delete`
- B.** When a local object reaches the end of its scope, or is deleted
- C.** At the end of `main()` , for every object
- D.** When the garbage collector decides

The Copy That Bites

```
PyList a;  
a.append(3);  
a.append(7);
```

```
PyList b = a;           // what exactly got copied?
```

a.myArray	0x5f2a10	→ [3 7]
b.myArray	0x5f2a10	→ the same block!

The default copy duplicates the **pointer**, not the data.
Two objects now own one block, a **shallow copy**.

TALK TO YOUR NEIGHBOR · TTYN

What goes wrong with that shallow copy?

- A.** Nothing: sharing the block saves memory
- B.** Changing `b[0]` also changes `a[0]`
- C.** Both destructors delete the same block, undefined behavior
- D.** Both B and C

Deep Copy: the Copy Constructor

```
PyList::PyList(const PyList &orig) {  
    mySize      = orig.mySize;  
    myCapacity  = orig.myCapacity;  
    myArray     = new Item[myCapacity];    // our OWN block  
    for (unsigned i = 0; i < mySize; ++i) {  
        myArray[i] = orig.myArray[i];      // copy the values  
    }  
}
```

Same shape as the growth code: allocate, copy, own. Now `a` and `b` are independent, and each destructor frees its own block.

Three Times It Copies

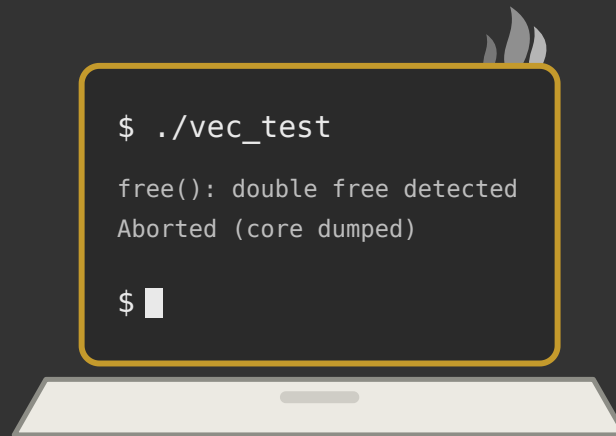
```
PyList b = a;                                // 1. explicit copy

void show(PyList v);                          // 2. pass-by-value
show(a);

PyList makeList() {                           // 3. return by value
    PyList local;
    return local;
}
```

Case 2 is why week 1 said *pass big objects by const reference*. Every by-value call to `show()` copies the whole heap block.

Demo



Watching a shallow copy destroy itself.

(two objects, one array, zero survivors)

Returning a Reference: `operator[]`

```
Item& PyList::operator[](unsigned i) {  
    return myArray[i];           // a reference, not a copy  
}  
  
v[2] = 99;           // works because v[2] IS the slot
```

Return `Item` and you get a copy, assigning to it would change a temporary and vanish. Returning `Item&` makes the call an *alias* for the element.

The `this` Pointer

Python

```
def set_first(self, v):  
    self.my_first = v
```

C++

```
void setFirst(Item v) {  
    this->myFirst = v;  
}
```

C++ passes the object silently as `this` : a pointer to the object the method was called on. You rarely write it, but it's there, and `this->x` is just `(*this).x`.

Week 04 Recap

- `new` gets heap memory at runtime; every `new` needs one `delete`
- **size** vs **capacity** is what makes append cheap most of the time
- Growing means allocate, copy, delete, repoint: linear, so we double
- If your class calls `new`, it needs a **destructor**
- The default copy is **shallow**: aliasing plus a double delete
- Write a copy constructor that allocates its own block

Next: a05 →

